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Chien-Hung Liu, Wen-Yuh Jywe, Cheng-Chung Hsu, et al.



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Development of a laser-based high-precision six-degrees-of-freedom motion errors measuring system for linear stage

Chien-Hung Liu^{a)}

Institute of Electro-Optical and Materials Science, National Formosa University, Huwei, 632, Yunlin, Taiwan, Republic of China

Wen-Yuh Jywe^{b)}

Department of Automation Engineering, National Formosa University, Huwei, Yunlin, Taiwan, Republic of China, 632

Cheng-Chung Hsu

Institute of Mechatronic Systems, Chien Kuo University of Technology, Changhua, Taiwan, Republic of China, 500

Tung-Hui Hsu

Department of Mechanical Engineering, National Chung Cheng University, San-Hsing, Min-Hsiung, Chia-Yi 621 Taiwan, Republic of China

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This article presents a useful measuring system for the simultaneous measurement of six-degrees-of-freedom motion errors of a moving stage. The system integrates a miniature fiber coupled laser interferometer with specially designed optical paths and quadrant detectors, capable of measuring six-degrees-of-freedom motion errors. Using this model, the proposed measuring method provides rapid performance, simplicity of setup, and preprocess verification of a linear stage. The experimental setups and measuring procedures, and a systematic calculated method for the error verification are presented in the paper. The system's resolution of measuring straightness error component is about 25 nm. The resolution of measuring the pitch and yaw angular error component is about 0.06 arcs. With the comparison between the HP calibration system and the proposed system in the measuring range of 120 mm, the system accuracy of measuring straightness error and angular error is within the range $\pm 0.6 \mu\text{m}$ and ± 0.3 arcs. © 2005 American Institute of Physics.

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I. INTRODUCTION

Recently, high accuracy stages have commonly been used in semiconductor processing, printed circuit board (PCB) drilling processing, micromachining processing, precision assembly processing, inspection processing, etc. The motion stage in a plane will have position and orientation errors in six degrees of freedom, which are three linear errors and three rotational errors. The characteristics of the motion stage will have a serious affect on the accuracy of machining processing and the quality of the machined product. The high precision six-degrees-of-freedom measuring system is important to measure and control the pose of an object on a motion stage with high accuracy. Current trends require a multi-degrees-of-freedom measuring system that can simultaneously on-line monitor the linear displacement and rotation displacement of an object along three orthogonal axes.¹⁻⁴ Alternatively, the use of off-line error compensation techniques has been recognized as an effective way to improve the performance of the motion stage.⁵⁻¹¹ The laser interferometer system is the most representative instrument for the measurement of displacement, straightness, pitch, and

yaw angular errors. It is considered to be the most reliable and accurate parametric calibration instrument. However, with one setup, the laser interferometer system can only measure one error element, and the optic setup and laser-path alignment is time consuming before every measurement work. Thus, some new methods and measuring systems have been developed to measure multiple geometric errors simultaneously to speed up the calibration. The following methods have been developed for on-line or off-line measurement using multiple laser interferometers, multiple plane mirrors, multiple laser beams, multiple prisms, and multiple quadrant detectors. Sommargren¹ used a dual-laser interferometer measurement system to measure linear and angular displacement simultaneously for wafer stage. Nakamura *et al.*² used four laser interferometers and a corner cube to measure the spatial coordinate of a microscopic scanning stage. Also, Lee *et al.*³ used two laser interferometers for linear displacement measurement along the X and Y axes and an angular measuring interferometer for the yaw angular error measurement. Fang *et al.*⁴ developed a six-degrees-of-freedom measurement system using four laser Doppler scales and two quadrant detectors to measure three linear displacement errors and three rotational errors simultaneously. The measuring systems described above are developed for on-line measurement of an X-Y stage. For off-line error compensation techniques,

^{a)}Electronic mail: carus@nfu.edu.tw

^{b)}Electronic mail: jywe@nfu.edu.tw

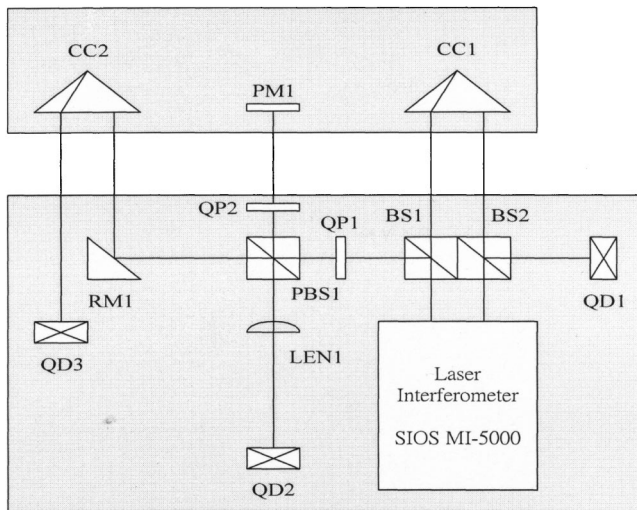


FIG. 1. The optical layout of the integrated laser based six-degrees-of-freedom measuring system.

several measuring systems have been proposed to increase the number of error components measured at a single setup for error calibration. The optical methods using a laser interferometer, multiple laser beams, and multiple quadrant detectors were developed for simultaneously measuring the multiple errors in a particular setup of these measurement devices.⁵⁻⁸ The other method⁹ used multi-charge coupled devices to scan the target which multispecific spots were installed on. The target attitude can be calculated from the positions of the specific spots in the coordinate. Kim *et al.*¹⁰ successfully developed a six-degrees-of-freedom measuring system using a circle grating target and three quadrant detectors. Three detectors detect the deflection angle of +1, 0, -1-order diffracted rays to calculate the six-degrees-of-freedom displacements. Jywe *et al.*¹¹ developed an integrated measuring system using two one-dimensional position silicon detectors. This integrated system can measure the geometric errors and dynamic contouring test.

This article proposes an integrated fiber coupled laser interferometer measuring system, capable of measuring the six-degrees-of-freedom motion errors. It utilizes a miniature

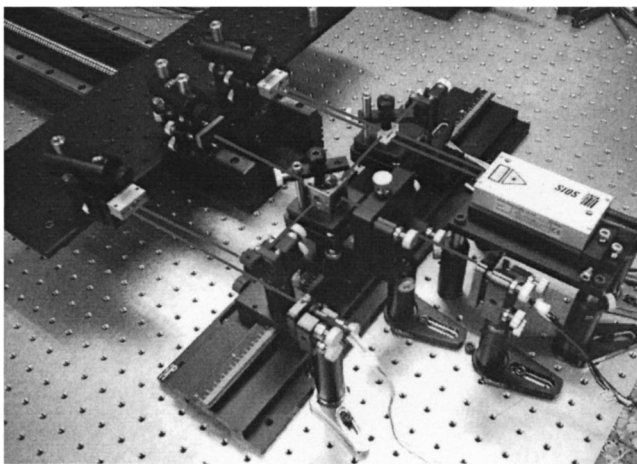


FIG. 2. Photograph of the proposed laser based six-degrees-of-freedom measuring system.

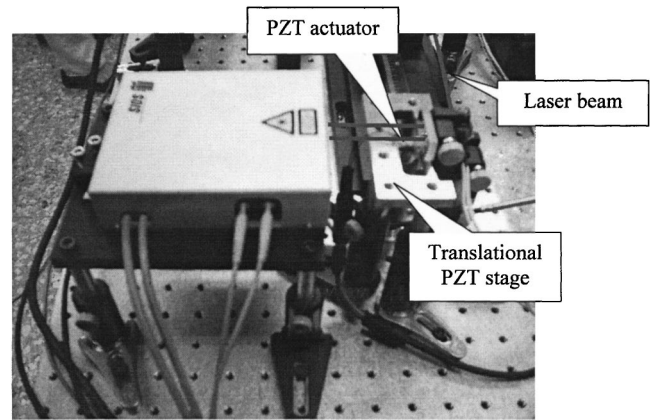


FIG. 3. The experimental setup for determining the resolution of displacement.

fiber coupled laser interferometer with some special optic setup and laser-path design. The miniature fiber coupled laser interferometer integrates the beam splitter and the detector of the interference fringe into one part. Thus, it reduces the difficulty of the laser-path alignment procedure and can be easily expanded to have the additional measuring function of simultaneously measuring the six-degrees-of-freedom motion errors. Current trends require designing a precision stage that can simultaneously compensate for the six-degrees-of-freedom motion errors to reach the submicron scale or nanometer scale within a long traveling range. Thus, the proposed system can be applied to the precision position and orientation control for a long traveling stage.

II. OPTICAL LAYOUT OF THE INTEGRATED LASER-BASED SIX-DEGREES-OF-FREEDOM MEASURING SYSTEM

A. Overall system layout

Figure 1 shows the simple layout of the integrated six-degrees-of-freedom measuring system. The system is composed of two parts: A moving part and a fixed part. The moving part is composed of two corner cubes (C1 and C2) and a plane mirror (M1). The fixed part is composed of a miniature fiber couple laser interferometer, two beam splitters (BS1 and BS2), and a polarizing beam splitter (PBS), two quarter-wave plate and three quadrant detectors (QD1, QD2, and QD3). First, the laser beam of the interferometer passing through the BS1 is divided into two laser beams. The passing beam is reflected by a corner cube (C1) and is di-

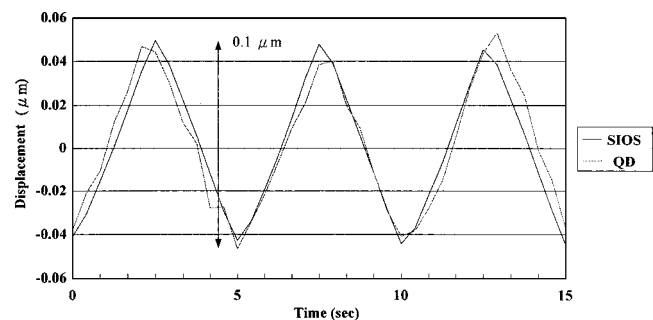


FIG. 4. The first test result of the displacement resolution of the QD.

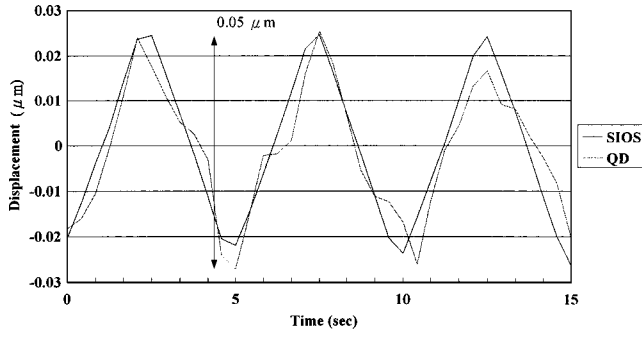


FIG. 5. The second test result of the displacement resolution of the QD.

vided into two laser beams again by BS2. One beam passes into the laser interferometer and the other one is projected on to QD1 to measure the vertical and horizontal straightness errors. The beam reflected by BS1 passes through a quarter-wave plate (QP1) and a PBS. The direction of polarization of the laser beam lies in a plane neither parallel nor perpendicular to the incident plane of the PBS, and thus results in one portion of the beam (namely, the P-polarized beam) passing through the PBS while the rest (namely, the S-polarized beam) is reflected. The beam reflected from the PBS passes through a quarter-wave plate (QP2) and then is reflected by a mirror (M1). The reflected beam passes through the quarter-wave plate and the PBS again and projects on to QD2 to measure the pitch and yaw angular errors. The beam passing through the PBS (P-polarized beam) is reflected by a reflector and a corner cube (C2) and projected on to QD3 to measure the error induced by roll angular error. A photograph of the proposed measuring system is shown in Fig. 2. Due to the design of the optical paths, the six-degrees-of-freedom motion errors can be measured simultaneously.

III. MEASURING METHOD OF THE INTEGRATED SIX-DEGREES-OF-FREEDOM MEASURING SYSTEM

A. Measurement of positioning error

The resolution of the laser interferometer is 1.24 nm and the accuracy is 0.05 ppm. The positioning error of a linear stage along its motion axis is measured by direct comparison of the displacement of the stage and the reading of the laser interferometer. That is

$$\text{Positioning error } (\delta_x) = L - L_1, \quad (1)$$

where L is the measuring reading of the laser interferometer, and L_1 is the displacement of the stage.

B. Measurement of the yaw and pitch angular errors

The measurement principle of the yaw and pitch angular errors is based on the autocollimator. As shown in Fig. 1, the beam reflected by a mirror (M1) and projected on to the QD₂ is used to measure the pitch and yaw angular errors. The QD₂ is placed apart from the focal point of the lens so that a light spot with a width of w is generated on the QD₂. The position of the light spot can be calculated by:¹²

$$X_{QD2} = \frac{(I_1 + I_4) - (I_2 + I_3)}{I_1 + I_2 + I_3 + I_4} \times 100\% = \frac{2}{w} \Delta x, \quad (2)$$

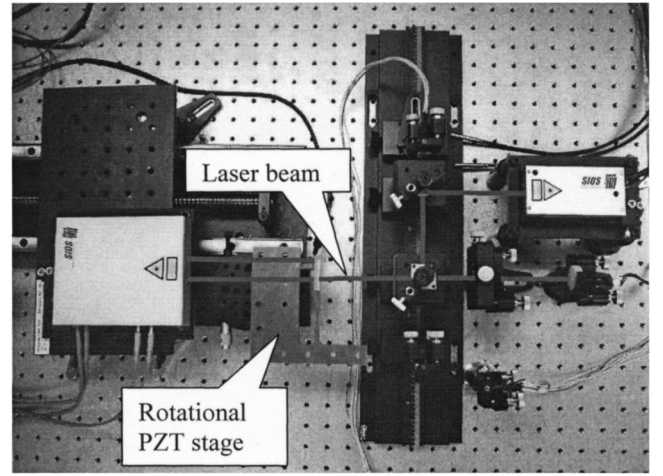


FIG. 6. The experimental setup for determining the rotational resolution.

$$Y_{QD2} = \frac{(I_1 + I_2) - (I_3 + I_4)}{I_1 + I_2 + I_3 + I_4} \times 100\% = \frac{2}{w} \Delta y, \quad (3)$$

where I_1 , I_2 , I_3 , and I_4 are the photoelectric currents from the four quadrants of QD₂. Then, the yaw and pitch angular error can be calculated by

$$\text{Yaw angular error: } \varepsilon_z = \frac{w X_{QD2}}{4f} = k X_{QD2}, \quad (4)$$

$$\text{Pitch angular error: } \varepsilon_y = \frac{w Y_{QD2}}{4f} = k Y_{QD2}. \quad (5)$$

It can be seen that by adjusting the location of the QD₂ relative to the focal position of the lens, the proper measurement range/sensitivity and resolution can be obtained.

C. Measurement of the straightness and roll angular errors

As shown in Fig. 1, the corner cube (C1) is used to measure the horizontal and vertical straightness errors of the moving stage. It is known that when the corner cube has a horizontal and vertical displacement, the laser beam reflected by the corner cube will magnify twice the displacement along the x and y direction of the QD₁. Thus, the horizontal and vertical straightness can be calculated by

$$\text{Horizontal straightness: } \Delta_y = \frac{X_{QD1}}{2}, \quad (6)$$

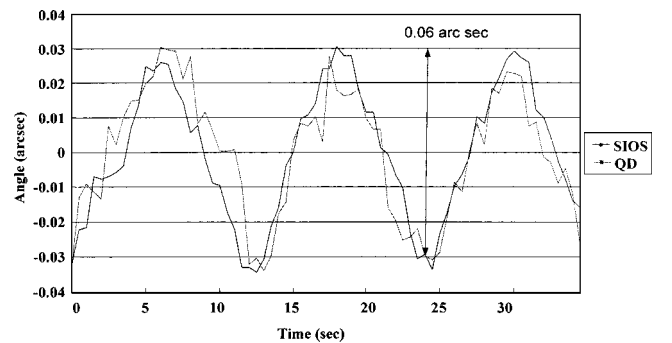


FIG. 7. The test result of the angular resolution.

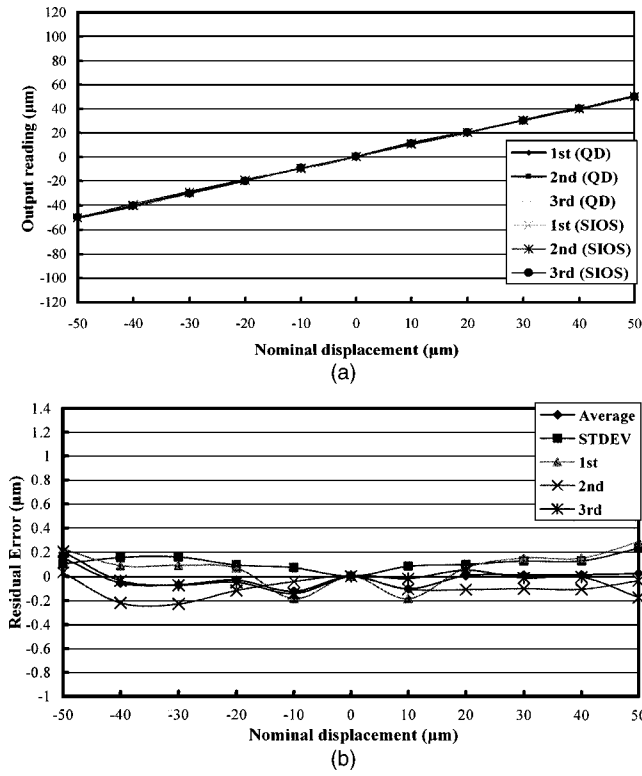


FIG. 8. (a) The calibration result of linearity of the QD. (b) The residual error.

$$\text{Vertical straightness: } \Delta_z = \frac{Y_{QD1}}{2}, \quad (7)$$

where (x_{QD1}, y_{QD1}) is the output of QD1.

The roll angular error is measured by comparing the output in the vertical direction (y direction) of QD1 and QD3. That is

$$\text{Roll angular error: } \varepsilon_x = \frac{\frac{Y_{QD3}}{2} - \frac{Y_{QD1}}{2}}{d} = \frac{Y_{QD3} - Y_{QD1}}{2d}, \quad (8)$$

where d is the separation distance between the two corner cubes.

IV. DISCUSSION

In order to verify the proposed system, two lead-zirconate titanate (PZT) stages (one is the translational stage and the other is the tilt angular stage) were developed for the precision verification of the proposed system. The measuring setups were all set up on an optical table. A dual-beam laser interferometer with a displacement resolution of 1.24 nm and an angular resolution of 0.01 arcsec was used as the reference to verify the proposed system.

A. Determining the resolution for straightness measurement

The experimental setup of the displacement resolution test is shown in Fig. 3. One plane mirror and the test QD were mounted on the translational test stage driven by PZT

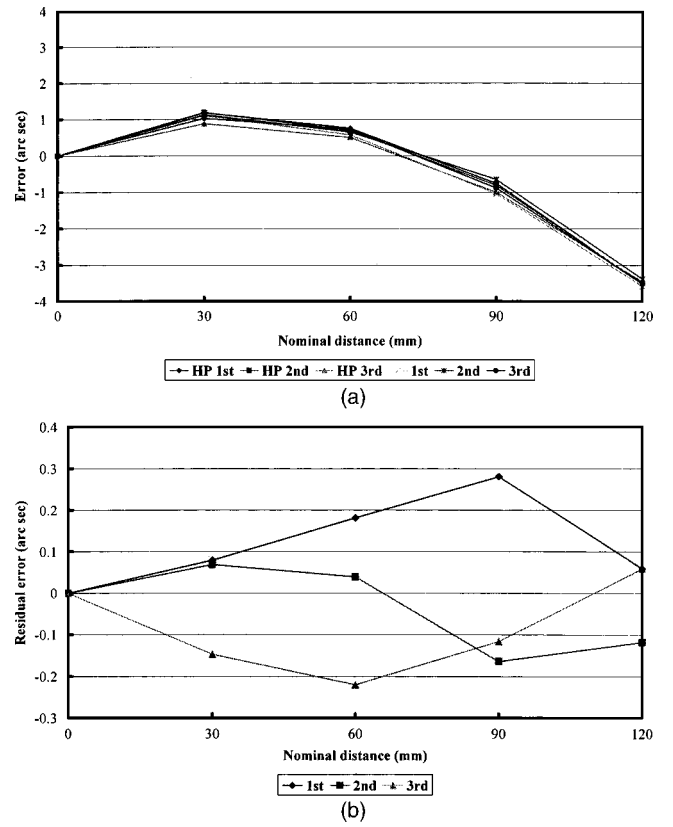


FIG. 9. (a) Comparison results of pitch angular errors between HP calibration system and the proposed system. (b) Residual errors between two calibration systems.

to reflect the laser beam from the laser interferometer. The triangulation voltage signal was applied to the PZT driver of the stage to move the QD and plane mirror. The laser interferometer is used as a reference to check the resolution of the QD. Figures 4 and 5 show some test results of the QD. Both of the outputs with respect to time are plotted in Figs. 4 and 5. As shown in Fig. 4, the output of the laser interferometer showed an amplitude of 1 μm very well, and the QD also responded to the 1 μm quite well. As also shown in Fig. 5, the output of the laser interferometer showed an amplitude of 0.05 μm very well, and the QD also show a similar response to the amplitude of 0.05 μm. The resolution of the QD is estimated about 0.05 μm. The resolution of straightness measurement can be calculated as

$$R_{\text{straightness}} = \left[\frac{\left(\frac{\text{Resolution of QD}}{2} \right)^2}{2} \right]^{1/2}. \quad (9)$$

Thus, the resolution of straightness can be calculated as 0.025 μm.

TABLE I. Estimation of system error within the range of ±50 μm and ±50 arcs.

Measurement element	Estimated system error
$d\varepsilon_z$	±0.28 arcs
$d\varepsilon_y$	±0.28 arcs
$d\varepsilon_x$	±0.2 arcs
$d(\Delta_x)$	±0.25 μm
$d(\Delta_y)$	±0.25 μm

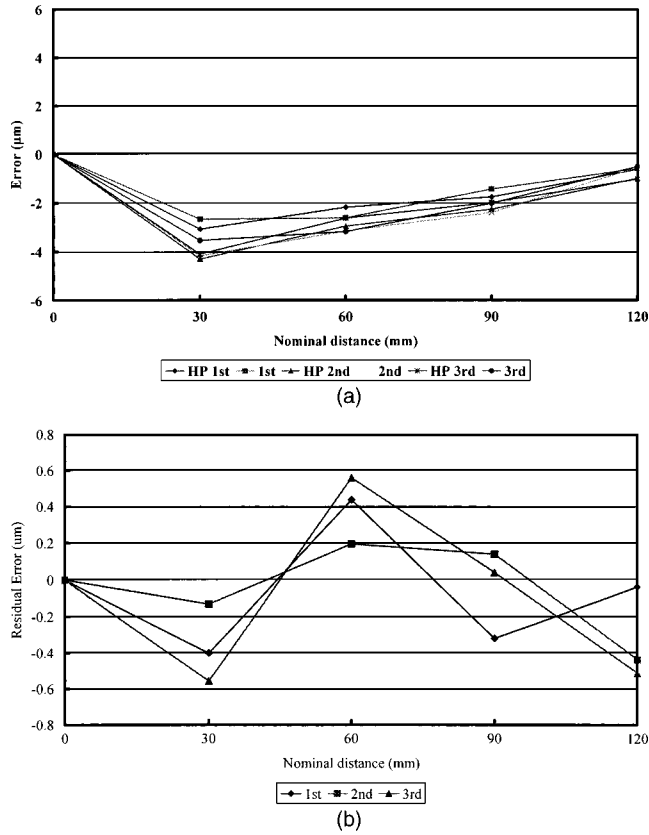


FIG. 10. (a) Comparison results of vertical straightness errors between HP calibration system and the proposed system. (b) Residual errors between two calibration systems.

B. Determining the resolution of the pitch, yaw, and roll angular errors

The experimental setup of the angular resolution test is shown in Fig. 6. Two plane mirrors were mounted on the tilt angular test stage. One was for the dual-beam laser interferometer and the other was for the proposed angular measuring system. The position of the QD on the optical axis was carefully adjusted to get high sensitivity of angle detection. The test result is shown in Fig. 7. The result shows that the output amplitude of the QD is similar to the amplitude of the dual-beam laser interferometer. The resolution is approximately 0.06 arcs.

The roll angular resolution can be calculated as

$$R_{\text{roll}} = \left[\sum \left(\frac{\text{resolution of QD}}{2d} \right)^2 + \left(\frac{\text{resolution of QD}}{2d} \right)^2 \right]^{1/2}, \quad (10)$$

where $d = 200.112$ mm. Thus, the resolution of roll measurement can be calculated as 0.02 arcs.

C. Calibration test of quadrant detector

The calibration tests of the QD were carried out with an SIOS laser interferometer for three runs. The accuracy calibration of the positioning errors was made in direct comparison with the laser interferometer. The calibration test results of the QD are shown in Fig. 8. The residual error of the QD

was found to be within $\pm 0.28 \mu\text{m}$ for a measuring range of $\pm 50 \mu\text{m}$. Also, the repeatability analysis result was shown in Fig. 8. The repeatability test is $\pm 0.11 \mu\text{m}$.

D. Estimate of the system error

In order to get the inaccuracy of the proposed measuring system, every measuring element is estimated using partial differentiation. Suppose that every measuring element F_i is a function of a number of individual signals, s_i , from the QD and positional parameters. The compound error of F_i can be expressed as

$$dF_i = \sum \left[\frac{\partial F_i}{\partial s_i} \delta s_i \right], \quad (11)$$

where dF_i is the accuracy of the measuring element F_i . The positional parameter, d , can be easily measured to an accuracy of $\pm 10 \mu\text{m}$. Thus, the accuracy of each measuring term can be derived from Eqs. (4)–(8) and can be represented as

$$d\epsilon_z = \pm k \delta X_{\text{QD}}, \quad (12)$$

$$d\epsilon_y = \pm k \delta Y_{\text{QD}}, \quad (13)$$

$$d(\Delta_x) = \pm \frac{1}{2} \delta X_{\text{QD}}, \quad (14)$$

$$d(\Delta_y) = \pm \frac{1}{2} \delta Y_{\text{QD}}, \quad (15)$$

$$d\epsilon_x = \pm \left[\frac{1}{2d} (\delta Y_{\text{QD3}} - \delta Y_{\text{QD1}}) + \frac{Y_{\text{QD1}} - Y_{\text{QD3}}}{2d^2} \delta d \right]. \quad (16)$$

According to the above equations, the estimated compound errors for every measuring item are shown in Table I.

E. Verification results of the proposed system

The proposed system was verified by the traditional calibration system (HP calibration system) simultaneously using the same stage. Measurements were performed by moving the stage along each axis for a distance of 120 mm with an increment of 30 mm, for three runs. The system measures and calculates via the outputs of each of the quadrant detectors and also compares to the HP calibration system. (The figures only show the measurement results of vertical straightness errors and pitch angular errors). The verification results were evaluated and showed in Figs. 9 and 10. As shown in Fig. 9, the residual errors of the pitch angular error between two systems are calculated within the range of ± 0.3 arcs. The calculated result of the vertical straightness error in Fig. 10 shows the residual errors between both of the two systems is within the range of $\pm 0.6 \mu\text{m}$.

This article presents a useful measuring system for the simultaneous measurement of six-degrees-of-freedom motion errors of linear stage. In future work, the miniature procedure for the entire measurement elements will be studied.

ACKNOWLEDGMENT

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